WIP: Learning Adversarial Attacks on Adaptive Traffic Signal Control Systems Under Cooperative Perception

Wangzhi Li, Tianheng Zhu, and Yiheng Feng, *Purdue University*

https://www.usenix.org/conference/vehiclesec25/presentation/li-wangzhi

This paper is included in the Proceedings of the 3rd USENIX Symposium on Vehicle Security and Privacy.

August 11–12, 2025 • Seattle, WA, USA

978-1-939133-49-6

WIP: Learning Adversarial Attacks on Adaptive Traffic Signal Control Systems Under Cooperative Perception

Wangzhi Li
Purdue University

Tianheng Zhu
Purdue University

Yiheng Feng *
Purdue University

Abstract

Significant advancements in traffic control systems, such as integration with sensing and communication technologies, have led to increased system complexity. While these developments offer substantial benefits, they also introduce heightened vulnerabilities in cyberspace. This paper presents a security analysis of adaptive traffic control systems operating under cooperative perception environments with connected and automated vehicles (CAVs). To explore system vulnerabilities, we propose a novel reinforcement learning-based black-box adversarial attack framework, which demonstrates effectiveness against state-of-the-art adaptive traffic control systems. Specifically, the multi-action proximal policy optimization (multi-PPO) algorithm is employed to train the attacker agent capable of generating a fake CAV along with its "detected" vehicles. Experimental results indicate that the fake CAV can fool a learning-based traffic control system by injecting falsified detection data, leading to a 62.5% increase in average vehicle delay.

1 Introduction

Traffic signal control systems have experienced significant advancements over the past few decades in both data collection methods and control models. From the data collection's perspective, infrastructure-based sensors such as loop detectors are widely deployed in about half of the signalized intersections across U.S. However, they are limited by the fixed location as well as the high cost of installation and maintenance. Starting from the 2010s, connected vehicle (CV) technology has been proven to be more effective in collecting spatial-temporal vehicle trajectory information for traffic signal control [10, 23, 33]. One major limitation is that a critical CV market penetration rate (MPR) is required for real-time operations, usually at least 10%, which is not expected to be achieved soon [11, 32]. Recently, connected and automated vehicles (CAVs), equipped with advanced sensing, edge computing, and vehicle-to-everything (V2X) communication capabilities, bring a new paradigm in traffic data collection [5, 20]. Researchers have shown that CAV-based cooperative perception, which leverages onboard sensors and V2X communications to collaboratively collect traffic data with other vehicles and roadside infrastructure, can further improve the data collection efficiency [4, 8, 14] under a very lower MPR, as shown in Figure 1. Specifically, CAVs share the sensor perception data, using Sensor Data Sharing Message (SDSM) defined in SAE J3224 [26] where an SDSM contains the CAV's own information as well as its detected vehicles' information (e.g., location, speed, detection accuracy). From the signal control model perspective, new methods have been developed and tested, from traditional rule-based controllers (e.g., actuated signals), optimization-based controllers [7, 10], to recent learning-based controllers [1, 34]. In particular, recent research shows that deep reinforcement learning-based traffic signal control (DRL-TSC) systems have superiority and robustness against partial observation scenarios and have attracted more and more attention in the past decade [21, 23].

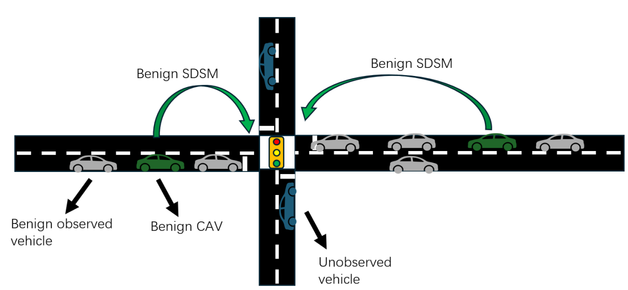


Figure 1: An illustration of cooperative perception.

Although the aforementioned new data collection and modeling approaches can bring significant benefits, the increasing traffic signal control system complexity, connectivity, and vulnerabilities in the control models can bring new threats. For

*Corresponding author. Address: 550 Stadium Mall Drive, West Lafayette, IN 47907. Email: feng333@purdue.edu

example, by strategically inserting only one falsified Basic Safety Message (BSM), Chen et al. [9] exploited a model-based adaptive traffic control system using CV trajectory data and reduced the traffic mobility to be 23.4% worse than without adopting the system. In terms of DRL-TSC systems, Haydari et al. [15] explored the attack and defense strategies. Two adversarial models, namely the Fast gradient Sign Method (FGSM) [13] and the Jacobian-based Saliency Map Attack (JSMA) [24], were developed to attack two typical DRL-TSC systems and generated a higher level of traffic congestion. However, in this study, the assumption that the attack can directly access the input pipeline of the DRL system can be hardly implemented in real-world scenarios. In addition, such an attack method requires knowledge of the target neural network parameters, i.e., a white-box attack, which is hard to realize with limited practicability. Qu et al. [25] investigated a scenario in which a group of malicious vehicles learn to compromise a 30-intersection real-world road network under DRL-TSC systems by sending fake vehicle presence information. However, this work assumes all colluding vehicles can access full traffic information over the whole network, which could be unrealistic.

Recently, research has been conducted on applying learning-based adversarial attacks, showing advantages in black-box attacks and in exploiting the weakness of the victim systems [12,27,35]. Irfan et al. [17] trained a deep Q-network (DQN) to learn to attack a waiting-time-based adaptive traffic signal controller. By inserting fake vehicles into approaches, the attacker successfully yields significant congestion. Similarly, Arabi et al. [2] explored training a DQN agent to attack an actuated TSC system. While previous works demonstrate the superiority of DRL-based attacks on TSC systems, they only target simple rule-based TSC systems and ignore the advanced optimization-based or learning-based TSC systems, which usually have much more complicated logic (e.g., state-action mapping) thus increasing the attack difficulty.

Limited studies have explored the security of adaptive TSC systems in a cooperative perception environment, especially under learning-based adversarial attacks. To fill this research gap, in this work, we propose a DRL-based attack model and perform a security analysis on cooperative perception-based adaptive TSC systems, including threat modeling, vulnerability profiling and impact analysis. We show the impacts of the proposed attack on one learning-based TSC system and one optimization-based TSC system through a simulation experiment with varying traffic demands and CAV penetration levels. In addition, we discuss a few potential detection and defense directions. The contributions of our work are summarized from the following three perspectives.

1. This is the first study, to the best of our knowledge, that provides a security analysis on cooperative perception-based adaptive TSC systems.

2. We propose a generalized learning-based black-box attack framework from target action selection to adversarial state construction, with the intention of increasing traffic congestion.

3. We experimentally demonstrate the effectiveness of our proposed attack method at a real-world intersection, considering realistic cooperative perception scenarios.

The rest of this paper is organized as follows. Section 2 introduces the target TSC systems, threat models, and attack methodologies. Section 3 presents the experiment settings and result analysis. Section 4 discusses the potential defense directions and future work of this WIP paper.

2 Methodology

In this section, we will first introduce the basic notions and terminologies in DRL, followed by the victim TSC systems, namely CAVLight and I-SIG. Based on this, we will introduce the learning-based attack framework.

2.1 Preliminaries: Deep Reinforcement Learning based Traffic Signal System

Reinforcement learning (RL) is a machine learning method for an agent to learn the optimal policy that can maximize the cumulative rewards by interacting with the environment. More details about RL can be found in [30]. In the context of traffic signal control, the signal controller is usually considered as the agent, and the traffic, including all road users, is considered as the environment. The mathematical formulation of RL is a Markov decision process (MDP) specified by a vector (S, A, R, Pr, γ). At a certain time t, the traffic controller agent observes a state $s \in S$, which describes the traffic environment (e.g., vehicle arrivals and current traffic signal status). Based on the state, the agent chooses one action $a \in A$, which can be the next traffic signal phase, and interacts with the traffic environment. In the next time step $t+1$, the state s transits to a new state $s' \in S$ and the agent receives a reward $r \in R$ which evaluates the performance of the previous actions, e.g., the reduction of all vehicles' delay. By interacting with the environment iteratively, the agent learns the optimal policy π^* maximizing the value functions such as $Q^\pi(s,a)$ (or $V^\pi(s)$) which estimates the future cumulative rewards at a given state s and action a (or at the given state s). Pr is the state transition probability, which describes the traffic evolving pattern and is usually unknown in real-world scenarios. γ is a discount factor used in the value function estimation. Deep reinforcement learning (DRL) further involves deep neural networks (DNNs) for function approximation, e.g. deep Q network (DQN) [22].

2.2 Target Adaptive Traffic Signal Control Systems

2.2.1 CAVLight

CAVLight is a cooperative perception-based DRL-TSC system [21], aiming at improving traffic control performance under the early deployment of CAVs.

When the CAV MPR is not 100%, the cooperative perception environment can only provide the CAVLight agent with partial information. In this case, the system is modeled as a partially observed MDP (POMDP) problem and denotes the observations by $o \in O$ where $O \subseteq S$. CAVLight uses an asymmetric advantage actor-critic algorithm (denoted as Asym-A2C) where the input of the actor is the estimated state based on the partial observation and the critic input is the ground truth state. By feeding the asymmetric input, the CAVLight agent learns to bridge the estimated state and the ground truth state. After the training process, the CAVLight agent can be deployed and solely requires observed data as input while showing robustness against the partial information scenario.

The CAVLight agent state, observation, action, and reward design are introduced below.

State. The state vector includes traffic status information and signal-related information. The traffic status information is composed of 1) the total number of vehicles in each segment of all incoming lanes controlled by a certain phase (in this study, all incoming lanes are separated into 3 equal-length segments. The protected left-turning lane is considered as one segment); and 2) the average speed difference of vehicles in each segment of each phase, where the average speed difference is defined as the difference between the average speed of vehicles and the free flow speed. For all components in the traffic-state-related input, an L2 norm is applied within each category to scale the input value. The traffic signal-related state includes 1) a one-hot vector to indicate the current traffic phase index and 2) the ratio of current phase duration over the maximum green time.

Observation. The observation shares the same format as the state but the information source only comes from the observed vehicles.

Action. Agent action comprises two options: switch or stay. Given a fixed phase sequence, at the end of one phase execution, the CAVLight agent chooses the next action. If the chosen action is to stay and the expected total duration does not exceed the maximum green time, the current phase continues for one more time step (1 second by default). Otherwise, the traffic signal switches to the next phase in the sequence, and a transition interval is added before the start of the next phase.

Reward. The reward function is defined as a weighted summation of two components: 1) the average delay of all vehicles in all incoming lanes, and 2) a phase-switching penalty which is a scaled value of the maximum waiting time of vehicles in the previous approach when the TSC agent makes the phase-changing decision.

2.2.2 I-SIG

I-SIG is first proposed by Feng et al. [10], stemming from the Controlled Optimization of Phase algorithm [29]. Dynamic programming is utilized to optimize phase duration and sequences, aiming at reducing total vehicle delay. Similar to CAVLight, in this work, the phase sequence is predefined and fixed. An arrival table is constructed based on the CAV cooperative perception result, indicating the predicted future arrival time and the number of arrival vehicles for each phase. The future arrival time of vehicles is calculated based on each observed vehicle's location and speed. At the beginning of each phase, I-SIG creates the arrival table based on the current observations from CAVs and optimizes the phase durations of a cycle accordingly.

2.3 Threat Model

For a given TSC system, the attacker tries to learn the optimal attack policy by interacting with the TSC and the traffic environment, as shown in Figure 2.

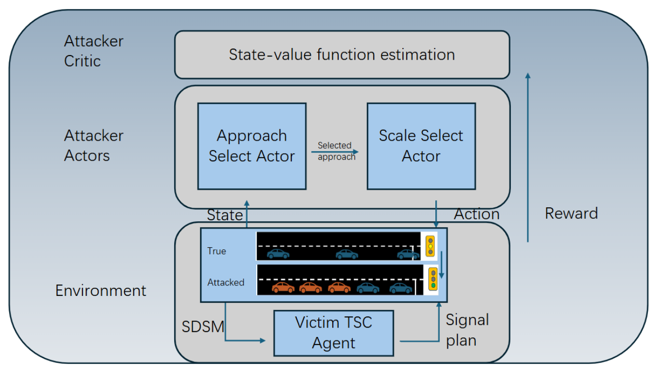


Figure 2: An illustration of the threat model.

The attacker can perform data spoofing attacks toward the CAV's perception or communication system by falsifying the detection or communication data. For example, the attack can compromise a CAV's perception sensors and generate fake objects [6] or compromise the onboard communication devices and send falsified messages [9]. Specifically, in this work, we assume that the attacker can only compromise one CAV's sensor or V2X communication device to generate and broadcast falsified cooperative perception messages in terms of SDSM. The falsified SDSM is received by the TSC system along with other benign SDSMs. The victim TSC system utilizes all received SDSMs to generate state/observation (for CAVLight) or arrival table (for I-SIG) to optimize traffic signals, as shown in Figure 3.

To realize such an attack, the attacker needs to determine 1) the target approach of the intersection and 2) the number of

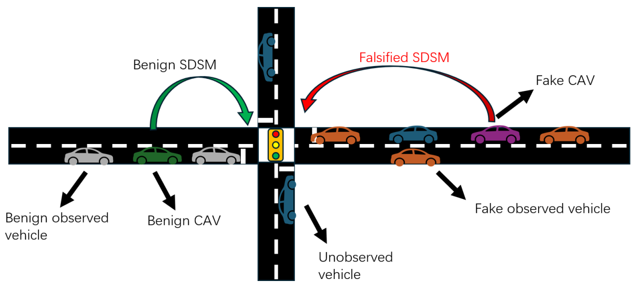


Figure 3: An illustration of data spoofing attack.

vehicles to be added in the fake SDSM. Due to the mechanism of cooperative perception, each SDSM is generated from one CAV. As a result, the added vehicles in the fake SDSM should be centered around an ego vehicle (i.e., the fake CAV), which should only be located in one approach of the intersection.

The process of learning adversarial policy is modeled as an MDP $(S_{att}, A_{att}, R_{att}, Pr_{att}, \gamma_{att})$ with a multi-action space [31], denoted as multiple MDPs (MMDP). In MMDP, the action set $A_{att} = A_{approach} \times A_{scale}$ is two-dimensional with two sub-action sets for the approach action and scale action, respectively. When the TSC agent is about to make a new decision, the adversary observes the traffic state and the victim TSC's status, $o_{att} \in O_{att}$ where $O_{att} \subseteq S_{att}$. Then it modifies the input state of the victim TSC system by inserting the fake CAV and its "detected" vehicles and determining the corresponding SDSM via $a_{approach} \in A_{approach}$ and $a_{scale} \in A_{scale}$. In this work, we assume that the attacker will generate a sequence of fake vehicles' information over a time period, imitating the real CAV cooperative perception scenario. The attack process is illustrated in Figure 4. The attacker generates the attack action 1 based on the observed state 1 at the decision-making time of the TSC agent. The attack action will be effective during the new phase and mislead the TSC agent to make the falsified decision at the next decision-making time. Therefore, the reward $r_{att} \in R_{att}$ of attacker action 1 is not fetched until the end of the execution of the falsified decision. In summary, the attacker's action targets the next TSC decision-making time and gets evaluated one signal phase after launching the attack.

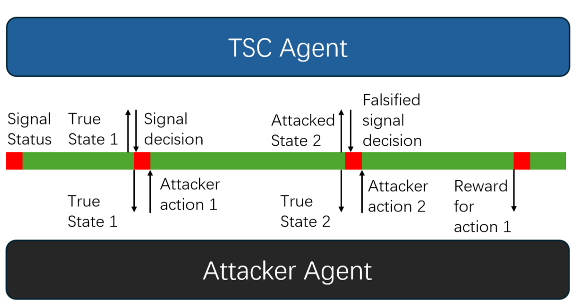


Figure 4: An illustration of the attack process.

We define the state, observation, action, and reward of the attacker as below.

State and observation. The attacker uses the same state and observation design as the victim CAVLight agent, including the number of vehicles, average speed, signal phase index, and phase duration.

Approach Action. The approach actor determines which approach the fake CAV will enter. The approach actor generates an array of probabilities that represent the chance of selecting the approach, respectively. It will then sample from the probability distribution and decide the final approach to add the fake CAV.

Scale Action. The scale action determines the number of surrounding vehicles to be added to each segment of the incoming approach. The scale actor will generate a series of numbers between 0 and 1 to represent the percentage of the maximum number of vehicles to be added to each segment. The maximum number of vehicles is a predefined constant during training and testing. For each through lane, we set three segments and for each left-turning lane, we set it as one separate segment. For example, if the final scale action is [4,3,2,1], it means 4 vehicles will be added to the first segment (closest to the stop bar), 3 vehicles to be added to the 2nd segment, 2 vehicles to be added to the 3rd segment, and 1 vehicle to be added to the left-turning lane segment.

Reward. The attacker's reward is a combination of the attack performance and the attack cost. We evaluate the attack performance using the total delay of all vehicles in the study area. The higher the delay the attacker causes, the more reward it receives. We also penalize the attacker by its scale action, i.e. the summation of total fake vehicles it added, to encourage stealthiness.

2.4 Multi-action Reinforcement Learning Attack Algorithm

In this work, we adopt the multi-action proximal policy optimization (multi-PPO) algorithm [19, 28, 31] as the training algorithm. One critic network with parameters ϕ is deployed to learn the state-value function $\hat{V}_\phi(s_{att,t})$, where the $\hat{V}_\phi(s_{att,t})$ is then used to compute the advantage function estimator $\hat{A}_t$ in Equation (1).

$$\hat{A}_t = \sum_{t'=t}^{T} \gamma_{att,t'} r_{att,t'} - \hat{V}_\phi(s_{att,t}) \tag{1}$$

Two actors' networks, θ_a for approach actor and θ_s for scale actor, learn the optimal policy, π_{θ_a} and π_{θ_s}, respectively. Noted here, in multi-PPO, we treat the two optimal policies π_{θ_a} and π_{θ_s} as two independent distributions. Therefore, we update each actor network separately by optimizing the clipped surrogate objective. For each of them, the clipped surrogate objective $L_{CLIP}(\theta)$ is calculated in Equation (2) and the actor networks are trained to maximize the $L_{CLIP}(\theta)$.

$$L_{CLIP}(\theta) = \hat{\mathbb{E}}_t[min\{\delta_t(\theta)\hat{A}_t, clip(\delta_t(\theta), 1-\varepsilon, 1+\varepsilon)\hat{A}_t\}] \quad (2)$$

where the $\delta_t(\theta)$ is the probability ratio between the update and the prior-update policy, which is defined as $\frac{\pi_\theta(a_{att,t|s_t})}{\pi_{\theta_{old}}(a_{att,t|s_t})}$. ε is a small hyperparameter that defines the clipping threshold.

The critic network is updated to minimize the Mean Square Error (MSE) as shown in Equation (3).

$$L_{MSE}(\phi) = \hat{\mathbb{E}}_t[MSE(r_{att,t}, \hat{V}_\phi(s_t))] \quad (3)$$

3 Experiments

3.1 Experiment Settings

A real-world intersection of Plymouth Road and Green Road in Ann Arbor, MI, USA, is used. The simulation is conducted in SUMO [3] simulation platform and the study area radius is 200 meters centering at the intersection. A 4-phase traffic signal timing plan is applied as illustrated in Figure 5, including main street straight and right turning (Main-THR), main street left turning (Main-L), side street southbound (Side-SB), and side street northbound (Side-NB) split phase. The phase sequence is Side-NB, Side-SB, Main-L, and Main-THR. For each phase, the minimum green time is 10 seconds (3 seconds for the Main-L), the maximum green time is 40 seconds, and the clearance interval includes a 4-second yellow and a 1-second all-red.

Figure 5: A real-world intersection of Plymouth Road and Green Road, Ann Arbor, MI, USA.

Each simulation round lasts 1,800 seconds, including a 100-second warm-up period, with a resolution of 0.1 seconds. Vehicle arrivals follow a Poisson process generated by SUMO. During training, traffic demand is set at 70% of real-world peak-hour demand, with an arrival rate of 2,662 vehicles per hour (vph). In deployment, demand varies dynamically: 100% (3,804 vph) for 0–600 seconds, 70% for 600–1200 seconds, and 120% (4,565 vph) for 1200–1800 seconds to mimic the demand fluctuations. Performance is assessed using average vehicle delay, with the mean and standard deviation over three test rounds. A 5% CAV penetration rate is applied in both training and testing. One fake CAV can be added to a selected approach and each segment in that approach can accommodate up to 10 additional fake "detected" vehicles.

Learning-related parameters of the CAVLight agent and its DNN structures can refer to [21] for more details as we adopt the same model from the previous work. As to the multi-PPO attacker, for all neural networks, three fully connected layers are used as hidden layers and the optimizer is Adam [18] with a 1e-3 learning rate. Each hidden layer consists of three times of input layer neurons. The He uniform [16] is adopted as the layer weight initializer for the hidden and output layers. The experience replay buffer size, batch size, discount rate, and iteration number used in this work are 5,000, 128, 0.99, and 5,000, respectively. Training loss are shown in Figure 9 in the Appendix, where the dashed line and numbers represent the average loss in the last 1,000 steps.

We adopt one random attacker as the benchmark algorithm, which is controlled by the initialized status attacker neural networks without any updates.

3.2 Result Analysis

We compare the performance of the CAVLight system under three conditions, without attack, with random attack, and with the proposed adversarial learning-based attack model, as shown in Table 1. The comparison result shows a significant increase on average delay under the proposed attack model.

Attacker	Average Delay (s)	STD	Increase (%)
None	45.6	1.6	/
Random	59.0	2.1	29.3%
Proposed	74.1	5.5	62.5%

Table 1: Comparison of average delay under different attack scenarios

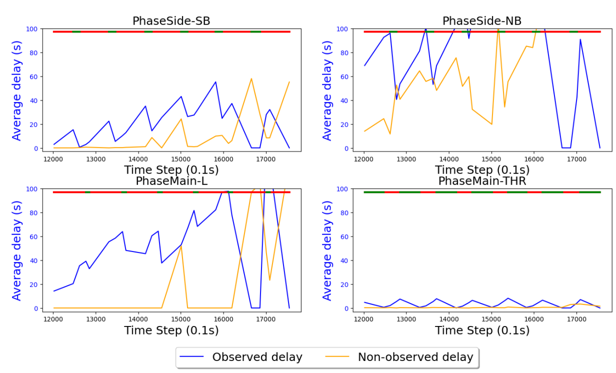


Figure 6: Vehicle average delay and signal phase under attack

We further visualize the vehicle average delay of each phase under the proposed attack model in Figure 6, and the delay in the normal case without attack in Figure 10 in the Appendix.

In each subfigure for a given phase, the x-axis is the simulation time and the y-axis is the average delay of all vehicles in oncoming lanes. The blue line represents the observed real vehicles' average delay (including CAVs and detected surrounding vehicles), and the orange line represents the average delay of unobserved vehicles. The green-red bar on the top of each sub-plot represents the signal phase status of the given phase. By comparing the attacked case and the normal one in this example, it can be observed that the attack creates more congestion on side streets and left-turning lanes on the main street by over-extending the Main-THR phase.

In order to better understand the attack strategy, we categorize the attacker's decision by the target phase (i.e., the attacker's approach action) under each current signal phase. The action distribution of the approach actor is shown in Figure 7. It can be seen that under all circumstances, the attacker tends to select main street approaches to insert fake vehicles. This will encourage the TSC agent to either extend the green phase on Main-THR or shorten the green time on the side street phases or Main-L, causing unnecessary long intervals of Main-THR or insufficient green time on other three phases, and thus causing higher delays.

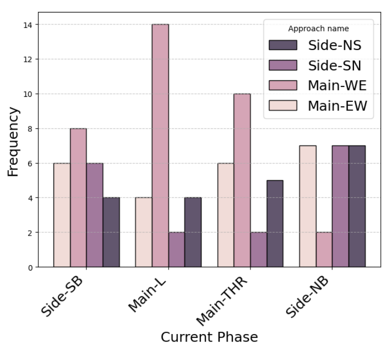


Figure 7: Approach actor's action distribution by current phase

In terms of the scale actor, its action distribution for a given current phase is visualized in Figure 8. Each subfigure represents the scale value distribution in the corresponding segment. From the figure, the scale actor tends to add the maximum number of fake vehicles into Segment 1 and Segment 2, which are the two segments closer to the stop bar. For Segment 3 and Segment left-turning, the scale actor tends to add much fewer fake vehicles. This can be explained by the fact that adding more vehicles closer to the stop bar results in a higher delay, thus leading the TSC agent to overextend the corresponding green phase.

4 Discussion and Future Work

As a WIP paper, we will continue to 1) develop the DRL attacker targeting the I-SIG TSC system. Based on this, we will

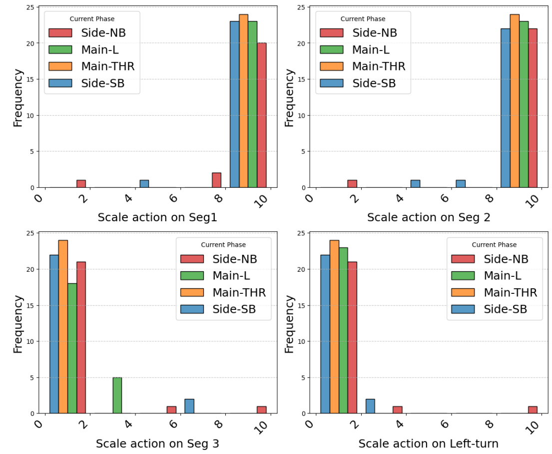
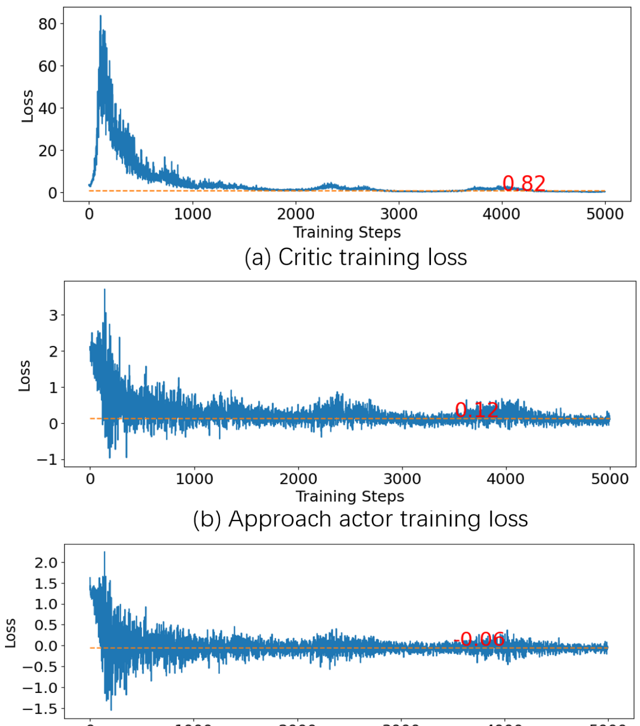


Figure 8: Scale actor's action distribution by current phase

further compare the difference in performance between the learning-based TSC and optimization-based TSC under the proposed attacks. 2) We will refine the multi-PPO algorithm to allow the attacker to explore and learn more diverse attack policies. Also, we will compare the performance of the proposed method with more benchmark attacker algorithms, for example, considering the attacks that create noise in communication and block communication periodically. 3) We will further consider the trajectory generation process of the fake vehicles in each SDSM to develop an end-to-end attack model. In other words, we will not only consider the final status of the fake vehicles at the decision-making time of the TSC systems but also collaboratively generate the trajectories of these fake vehicles when the fake CAV enters the communication range. 4) We will scale up the experiment to consider multiple intersections and investigate how the attacker can manipulate the fake SDSM to target a corridor or an urban network.

Although in this study, we do not consider detection and defense methods, some potential mitigation strategies are briefly discussed. The DRL attacker generates adversarial states without considering real traffic conditions (e.g., other vehicles' location and speed). As a result, the generated state may violate vehicle dynamics. Toward this end, the detection model can be designed based on physical laws and check whether the observations between two time steps violate physical boundaries or traffic flow properties. Certain consistency metrics can be developed to quantify the discrepancy and perform anomaly detection. As for the defense strategies, the simplest way is to fall back the controller to fixed-time or actuated control when attacks are detected. Moreover, since the (falsified) cooperative perception data requires a V2X communication network to be transmitted, revocation of the attacker's credentials is an effective defense method, if the compromised CAV can be identified correctly.

A Additional Figures



(a) Critic training loss

(b) Approach actor training loss

(c) Scale actor training loss

Figure 9: Training loss of attacker agent

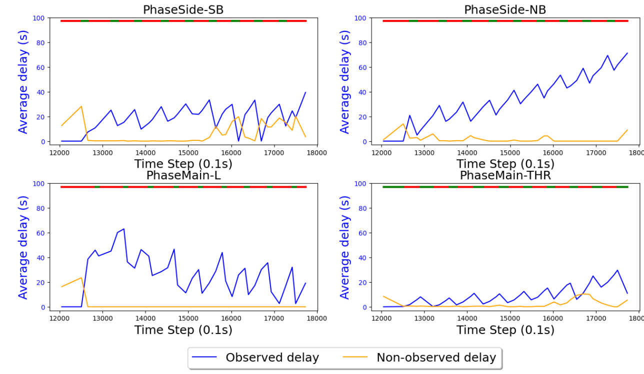


Figure 10: Vehicle average delay and signal phase under no attack

References

[1] Baher Abdulhai, Rob Pringle, and Grigoris J Karakoulas. Reinforcement learning for true adaptive traffic signal control. *Journal of Transportation Engineering*, 129(3):278–285, 2003.

[2] Najmeh Seifollahpour Arabi, Talal Halabi, and Mohammad Zulkernine. Reinforcement learning-driven attack on road traffic signal controllers. In *2021 IEEE International Conference on Cyber Security and Resilience (CSR)*, pages 218–225. IEEE, 2021.

[3] Michael Behrisch, Laura Bieker, Jakob Erdmann, and Daniel Krajzewicz. Sumo–simulation of urban mobility: an overview. In *Proceedings of SIMUL 2011, The Third International Conference on Advances in System Simulation*. ThinkMind, 2011.

[4] Antoine Caillot, Safa Ouerghi, Pascal Vasseur, Rémi Boutteau, and Yohan Dupuis. Survey on cooperative perception in an automotive context. *IEEE Transactions on Intelligent Transportation Systems*, 23(9):14204–14223, 2022.

[5] Peng Cao, Zhiqiang Xiong, and Xiaobo Liu. An analytical model for quantifying the efficiency of traffic-data collection using instrumented vehicles. *Transportation research part C: emerging technologies*, 136:103558, 2022.

[6] Yulong Cao, Chaowei Xiao, Benjamin Cyr, Yimeng Zhou, Won Park, Sara Rampazzi, Qi Alfred Chen, Kevin Fu, and Z Morley Mao. Adversarial sensor attack on lidar-based perception in autonomous driving. In *Proceedings of the 2019 ACM SIGSAC conference on computer and communications security*, pages 2267–2281, 2019.

[7] Tang-Hsien Chang and Jen-Ting Lin. Optimal signal timing for an oversaturated intersection. *Transportation Research Part B: Methodological*, 34(6):471–491, 2000.

[8] Hanlin Chen, Brian Liu, Xumiao Zhang, Feng Qian, Z Morley Mao, and Yiheng Feng. A cooperative perception environment for traffic operations and control. *arXiv preprint arXiv:2208.02792*, 2022.

[9] Qi Alfred Chen, Yucheng Yin, Yiheng Feng, Z Morley Mao, and Henry X Liu. Exposing congestion attack on emerging connected vehicle based traffic signal control. In *NDSS*, 2018.

[10] Yiheng Feng, K Larry Head, Shayan Khoshmagham, and Mehdi Zamanipour. A real-time adaptive signal control in a connected vehicle environment. *Transportation Research Part C: Emerging Technologies*, 55:460–473, 2015.

[11] Yiheng Feng, Jianfeng Zheng, and Henry X Liu. Real-time detector-free adaptive signal control with low penetration of connected vehicles. *Transportation Research Record*, 2672(18):35–44, 2018.

[12] Javier García, Rubén Majadas, and Fernando Fernández. Learning adversarial attack policies through multi-objective reinforcement learning. *Engineering Applications of Artificial Intelligence*, 96:104021, 2020.

[13] Ian J Goodfellow, Jonathon Shlens, and Christian Szegedy. Explaining and harnessing adversarial examples. *arXiv preprint arXiv:1412.6572*, 2014.

[14] Yi Guo and Jiaqi Ma. Drl-tp3: A learning and control framework for signalized intersections with mixed connected automated traffic. *Transportation Research Part C: Emerging Technologies*, 132:103416, 2021.

[15] Ammar Haydari, Michael Zhang, and Chen-Nee Chuah. Adversarial attacks and defense in deep reinforcement learning (drl)-based traffic signal controllers. *IEEE Open Journal of Intelligent Transportation Systems*, 2:402–416, 2021.

[16] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Delving deep into rectifiers: Surpassing human-level performance on imagenet classification. In *Proceedings of the IEEE international conference on computer vision*, pages 1026–1034, 2015.

[17] Muhammad Sami Irfan, Mizanur Rahman, Travis Atkison, Sagar Dasgupta, and Alexander Hainen. Reinforcement learning based cyberattack model for adaptive traffic signal controller in connected transportation systems. *arXiv preprint arXiv:2211.01845*, 2022.

[18] Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014.

[19] Kun Lei, Peng Guo, Wenchao Zhao, Yi Wang, Linmao Qian, Xiangyin Meng, and Liansheng Tang. A multi-action deep reinforcement learning framework for flexible job-shop scheduling problem. *Expert Systems with Applications*, 205:117796, 2022.

[20] Tao Li, Xu Han, and Jiaqi Ma. Cooperative perception for estimating and predicting microscopic traffic states to manage connected and automated traffic. *IEEE Transactions on Intelligent Transportation Systems*, 23(8):13694–13707, 2021.

[21] Wangzhi Li, Tianheng Zhu, and Yiheng Feng. A cooperative perception based adaptive signal control under early deployment of connected and automated vehicles. *Transportation Research Part C: Emerging Technologies*, 169:104860, 2024.

[22] Volodymyr Mnih, Koray Kavukcuoglu, David Silver, Andrei A Rusu, Joel Veness, Marc G Bellemare, Alex Graves, Martin Riedmiller, Andreas K Fidjeland, Georg Ostrovski, et al. Human-level control through deep reinforcement learning. *nature*, 518(7540):529–533, 2015.

[23] Zhaobin Mo, Wangzhi Li, Yongjie Fu, Kangrui Ruan, and Xuan Di. Cvlight: Decentralized learning for adaptive traffic signal control with connected vehicles. *Transportation research part C: emerging technologies*, 141:103728, 2022.

[24] Nicolas Papernot, Patrick McDaniel, Somesh Jha, Matt Fredrikson, Z Berkay Celik, and Ananthram Swami. The limitations of deep learning in adversarial settings. In *2016 IEEE European symposium on security and privacy (EuroS&P)*, pages 372–387. IEEE, 2016.

[25] Ao Qu, Yihong Tang, and Wei Ma. Adversarial attacks on deep reinforcement learning-based traffic signal control systems with colluding vehicles. *ACM Transactions on Intelligent Systems and Technology*, 14(6):1–22, 2023.

[26] SAE International. V2x sensor-sharing for cooperative and automated driving, 2022.

[27] Soumyendu Sarkar, Sajad Mousavi, Ashwin Ramesh Babu, Vineet Gundecha, Sahand Ghorbanpour, and Alexander K Shmakov. Measuring robustness with black-box adversarial attack using reinforcement learning. In *NeurIPS ML Safety Workshop*, 2022.

[28] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.

[29] Suvrajeet Sen and K Larry Head. Controlled optimization of phases at an intersection. *Transportation science*, 31(1):5–17, 1997.

[30] Richard S Sutton and Andrew G Barto. *Reinforcement learning: An introduction*. MIT press, 2018.

[31] Han Wang and Yang Yu. Exploring multi-action relationship in reinforcement learning. In *Pacific Rim International Conference on Artificial Intelligence*, pages 574–587. Springer, 2016.

[32] Qinzheng Wang, Yun Yuan, Xianfeng Terry Yang, and Zhitong Huang. Adaptive and multi-path progression signal control under connected vehicle environment. *Transportation Research Part C: Emerging Technologies*, 124:102965, 2021.

[33] Xingmin Wang, Zachary Jerome, Zihao Wang, Chenhao Zhang, Shengyin Shen, Vivek Vijaya Kumar, Fan Bai, Paul Krajewski, Danielle Deneau, Ahmad Jawad, et al. Traffic light optimization with low penetration rate vehicle trajectory data. *Nature communications*, 15(1):1306, 2024.

[34] Hua Wei, Guanjie Zheng, Huaxiu Yao, and Zhenhui Li. Intellilight: A reinforcement learning approach for intelligent traffic light control. In *Proceedings of the 24th ACM SIGKDD international conference on knowledge discovery & data mining*, pages 2496–2505, 2018.

[35] Xian Wu, Wenbo Guo, Hua Wei, and Xinyu Xing. Adversarial policy training against deep reinforcement learning. In *30th USENIX Security Symposium (USENIX Security 21)*, pages 1883–1900, 2021.